

Seminar 2 (8.10.2025)

Homogeneous relation: $\mathcal{R}: M \rightarrow M$

Equivalence relation:

- reflexive: $(\forall) x \in M, x \mathcal{R} x$
- transitive: $(\forall) x, y, z \in M, \left. \begin{array}{l} x \mathcal{R} y \\ y \mathcal{R} z \end{array} \right\} \Rightarrow x \mathcal{R} z$
- symmetric: $(\forall) x, y \in M, x \mathcal{R} y \Rightarrow y \mathcal{R} x$

Graph of a relation:

$$G = \{(x, y) \mid x \mathcal{R} y\}$$

Graph = set of ordered pairs

Partition:

$M_i \subseteq M, i \in \mathbb{N}^*$ for which $\left\{ \begin{array}{l} \bigcup_{i \in I} M_i = M \\ M_i \cap M_j = \emptyset, (\forall) i, j \in I, i \neq j \end{array} \right.$

$\rightarrow$ reunion of subsets
 $\hookrightarrow$ disjoint subsets

① $M = \{2, 3, 4, 5, 6\}$

$$x \mathcal{R}_1 y \Leftrightarrow x < y$$

$$x \mathcal{R}_2 y \Leftrightarrow x \mid y$$

$$x \mathcal{R}_3 y \Leftrightarrow \gcd(x, y) = 1$$

$$x \mathcal{R}_4 y \Leftrightarrow x \equiv y \pmod{3}$$

Write the graphs R, S, T, V .

$$R = \{(2, 3), (2, 4), (2, 5), (2, 6), (3, 4), (3, 5), (3, 6), (4, 5), (4, 6), (5, 6)\}$$

reflexive $\times$

transitive $\checkmark$

symmetric $\times$

Δ_M

$$S = \{(2,4), (2,6), (3,6), (2,2), (3,3), (4,4), (5,5), (6,6)\}$$

R ✓

T X

S X



all the elements in the set are in relation with each other.

$$T = \{(2,3), (2,5), (3,4), (3,5), (4,5), (5,6), (3,2), (5,2), (4,3), (5,3), (5,4), (6,5)\}$$

R X

T X

S ✓

$$V = \{(2,2), (2,5), (3,3), (3,6), (4,4), (5,5), (6,3), (6,6), (5,2)\}$$

R ✓

T ✓

S ✓

$$5 \equiv 2 \pmod{3}$$

$$5 : 3 = 1 \text{ r } 2$$

$$2 : 3 = 0 \text{ r } 2$$

$$\mathbb{Z}_m = \{\hat{0}, \hat{1}, \hat{2}, \dots, \hat{m-1}\}.$$

② $|A| = m$

$|B| = m$

a) The number of relations $f: A \rightarrow B$

b) The number of homogenous relations on A.

Sol.

a) $A \times B = \{(a,b) \mid a \in A, b \in B\}$

$|A \times B| = \text{card}(A \times B) = m \cdot m$

No. of relations: $2^{m \cdot m}$



(a,b) or (b,a)

b) $|A \times A| = m \cdot m = m^2$.

No of relations: 2^{m^2}

③ Give examples of relations :

→ Reflexive: $A = \{1, 2, 3\}$

$$G = \{(1,1), (2,2), (3,3)\} = \Delta_A$$

→ Transitive: $G = \{(1,2), (2,3), (1,3)\}$

→ Symmetric: $G = \{(1,2), (2,1)\}$

! A relation can be given by its graph.

④ $(\mathbb{R}, <)$

R ✓

T ✓

S x

$(\mathbb{N}, /)$

R ✓

T ✓

S x

$(\mathbb{Z}_1, /)$

R ✓

T ✓

S x

$(V^2, \perp)$

R x

T x

S ✓

lines
in space

$(V^2, \parallel)$

R x

T ✓

S ✓

$(V^3, \cong \Delta)$

R ✓

T ✓

→ congruence of triangles

~ Hr. total de fct:

$f: A \rightarrow B$

$$|B|^{|A|} = n^m$$

Hr. fct. bij:

$$m! = m!$$

Hr. fct. inj:

$$A_m^m$$

triangles

S ✓

$$(V^3, \cong \Delta) \begin{matrix} R \\ T \\ S \end{matrix} \begin{matrix} \checkmark \\ \checkmark \\ \checkmark \end{matrix}$$

→ similarity of triangles

(two lines d_1 and d_2 cannot be parallel to each other if they are equal ($d_1 = d_2$) because $d_1 \parallel d_2$ only if they have no points in common and $d_1 = d_2$ have all points in common)

⑤ $M = \{1, 2, 3, 4\}$

$R_1, R_2 \rightarrow$ homogenous relations on M .

$$R_1 = \Delta_M \cup \{(1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3, 2)\}$$

$$R_2 = \Delta_M \cup \{(1, 2), (1, 3)\}$$

π_1, π_2 partitions.

$$\pi_1 = \{\{1\}, \{2\}, \{3, 4\}\}$$

$$\pi_2 = \{\{1\}, \{1, 2\}, \{3, 4\}\}$$

a) $R_1, R_2 \rightarrow$ equivalent relations? Write the partition.

b) $\pi_1, \pi_2 \rightarrow$ partitions? Write the graph.

Sol.

a) R_1 -symmetrical because $(1, 2), (2, 1)$
 $(1, 3), (3, 1)$
 $(2, 3), (3, 2)$

- transitive because $(1, 2), (2, 3), (1, 3)$

- reflexive because Δ_M .

$\Rightarrow R_1$ equivalent relation. ; $\pi_{R_1} = \{\{1, 2, 3\}, \{4\}\}$

R_2 - not sym because there is no $(1, 2)$ and $(2, 1)$

- trans because $(\underline{1}, \underline{2}), (\underline{2}, \underline{2}), (\underline{1}, \underline{2})$
 $(\underline{1}, \underline{1}), (\underline{1}, \underline{2}), (\underline{1}, \underline{2})$
 $(\underline{1}, \underline{1}), (\underline{1}, \underline{2}), (\underline{1}, \underline{3})$

- reflexive because ΔM .
 $\Rightarrow r_2$ equivalent relation, $\pi_{r_2} = \{\{1,2,3\}, \{4\}\}$

$$\left. \begin{array}{l} b) \bigcup_{i \in I} \pi_i = M \\ \{1\} \cap \{2\} = \emptyset \\ \{1\} \cap \{3,4\} = \emptyset \\ \{2\} \cap \{3,4\} = \emptyset \end{array} \right\} \Rightarrow \pi_1 \text{ partition of } M.$$

$$G_{\pi_1} = \{(1,1), (2,2), (3,4), (3,3), (4,3), (4,4)\}$$

$$\bigcup_{i \in I} \pi_i = M$$

$\{1\} \cap \{1,2\} \neq \emptyset \Rightarrow \pi_2$ not a partition of M .

$$\text{Ex: 1) } M = \{1, 2, \dots, 6\}$$

$$\pi = \{\{1,2\}, \{3,4\}, \{5\}\} \text{ not a partition}$$

because there is no $\{6\}$, at least in relationship with itself.

$$2) G = \{(1,1), (1,2), (2,2), (3,3), (4,4), (5,5), (6,6)\}$$

$$\hookrightarrow \pi_G = \{\{1,2\}, \{3\}, \{4\}, \{5\}, \{6\}\}$$

? can we determine the graph of π if π is not a partition?

$$u = " \Rightarrow (1,1), (2,2)$$

⑥ Define θ on $\mathbb{C}$.

$$7 \cdot 4 \cdot 3 \cdot 1 = 2 \cdot 1 = 1 \cdot 2 \cdot 1$$

$$z_1 \sim z_2 \Leftrightarrow |z_1| = |z_2|$$

$$z_1 \sim z_2 \Leftrightarrow \arg z_1 = \arg z_2 \text{ or } z_1 = z_2.$$

a) $\sim, \simeq \rightarrow$ equivalent relations?

b) the quotient set (partition) of $\mathbb{C}/\sim, \mathbb{C}/\simeq$.

Sol.

a) $\sim$

$$R: |z_1| = |z_2|, (\forall) z_1, z_2 \in \mathbb{C} \Rightarrow \text{ref. (1)}$$

$$T: (\forall) z_1, z_2, z_3 \in \mathbb{C} : \begin{matrix} |z_1| = |z_2| \\ |z_2| = |z_3| \end{matrix} \Rightarrow z_1 = z_3 \Rightarrow \text{tr. (2)}$$

$$S: (\forall) z_1, z_2 \in \mathbb{C} : |z_1| = |z_2| \Rightarrow |z_2| = |z_1| \Rightarrow \text{sym. (3)}$$

(1), (2), (3) $\Rightarrow \sim \rightarrow$ eq. relation.

$$\mathbb{C}/\sim = \{z \in \mathbb{C} \mid |z| = |\bar{z}|\} = \{C(0, |z|) \mid z \in \mathbb{C}\}$$

$\simeq$

$$R: (\forall) z_1, z_2 \in \mathbb{C} : \arg z_1 = \arg z_2 \Rightarrow \text{ref. (1)}$$

$$T: (\forall) z_1, z_2, z_3 \in \mathbb{C} : \begin{matrix} \arg z_1 = \arg z_2 \\ \arg z_2 = \arg z_3 \end{matrix} \Rightarrow \arg z_1 = \arg z_3 \Rightarrow \text{tr. (2)}$$

$$S: (\forall) z_1, z_2 \in \mathbb{C} \Rightarrow \arg z_1 = \arg z_2 \Rightarrow$$

$$\Rightarrow \arg z_2 = \arg z_1$$

$$\Rightarrow \mathbb{C}/\simeq = \{z \in \mathbb{C} \mid \arg z = \arg \bar{z}\}$$

$\mathbb{T}_m/\mathbb{Z}$ $\cap$

